

Australian GHI Clustering

1 Introduction

We apply self organising maps (SOM) to analyse solar irradiance in the Australian region.

2 Methods

2.1 Data

Solar irradiance data was sourced from the Himawari satellite (hosted on the NCI Thredds server), where daily integral of global horizontal irradiance (GHI) [MJm^{-2}] across the Australian region (latitude -44.5 to -10.0, longitude 112.0 to 156.3, both at a resolution of 0.02) from 2020 to 2025 was used. The data was linearly interpolated to the Australian Albers equal area projection (similar to in [3]) to give equal weighting to equatorial regions versus the south-most regions during training. During the interpolation, the data was also coarsened to a grid of 55 km by 55 km cells.

2.2 SOM Analysis

The entire dataset was then split into the 4 seasons (DJF, MAM, JJA, SON), and separate SOMs trained for each season. Locations with NAN values (primarily over the ocean) were dropped. The Torch SOM[2][1] package in Python was used. All maps used a hexagonal grid with grid sizes determined through a hyper-parameter search. The default learning rate (0.5) was used, which linearly decreased to zero over the 200 epochs of training. The initial neighbourhood radius was set to 3, greater than the size of the grid, and linearly decayed to 1 over training.

3 Results

Figure 5 shows the final nodes for each seasonal SOM. For summer, a 3 by 3 grid was chosen based on the low quantisation and topographic error (figure 2b). Similar search on grid sizes was done for each season, and a 3 by 3 grid was chosen for all seasons except autumn.

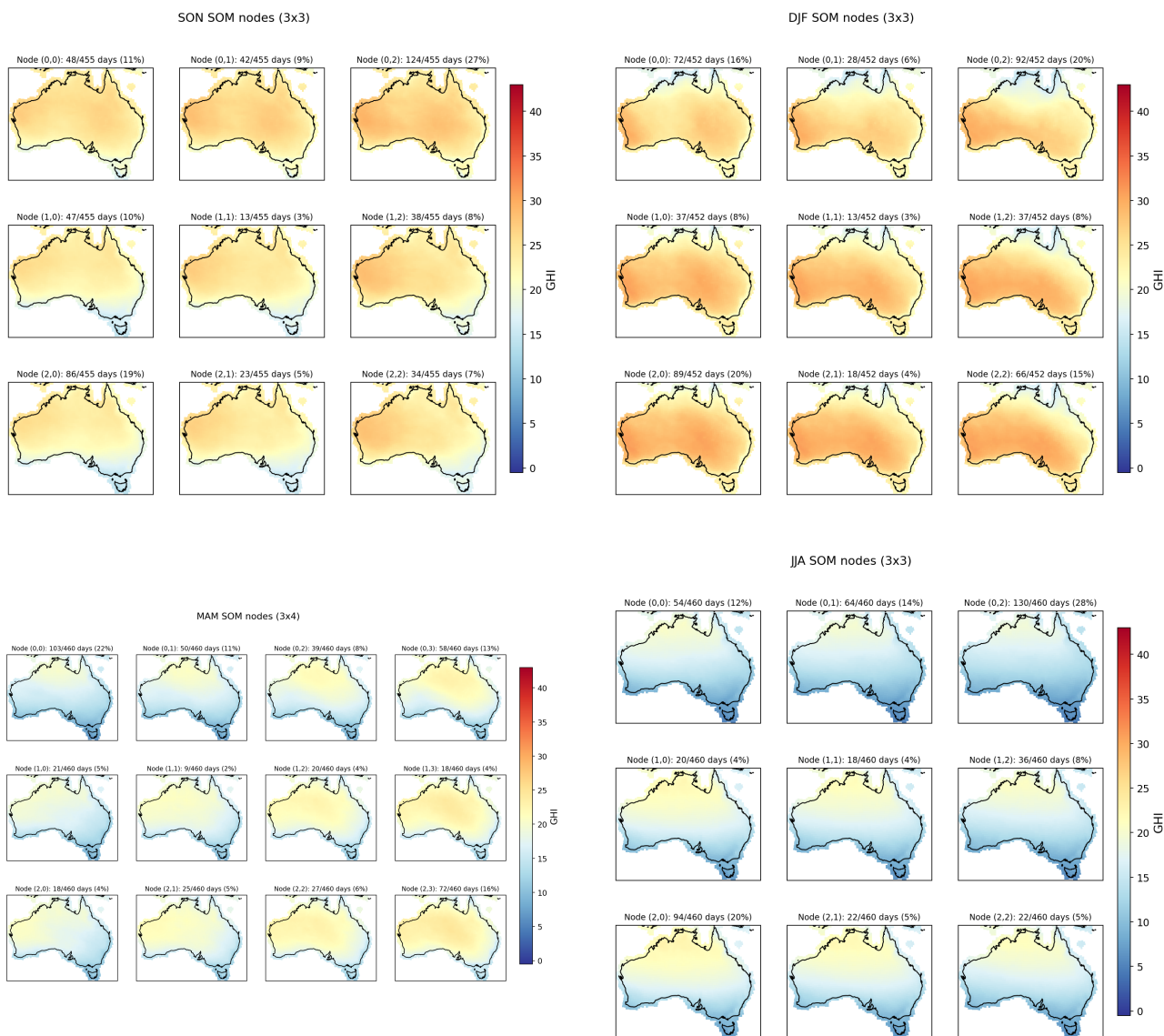
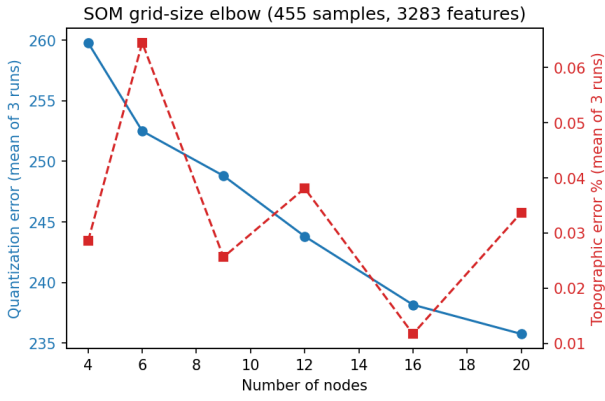
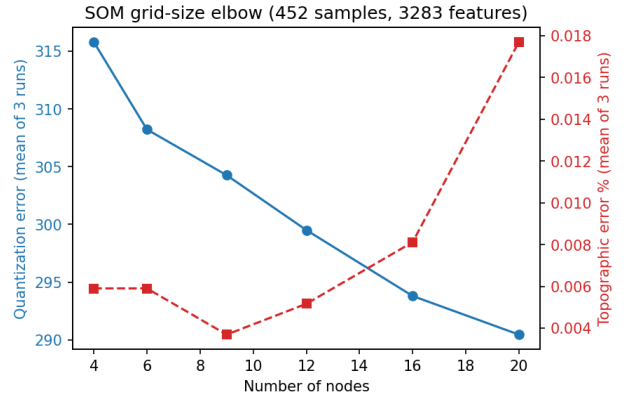


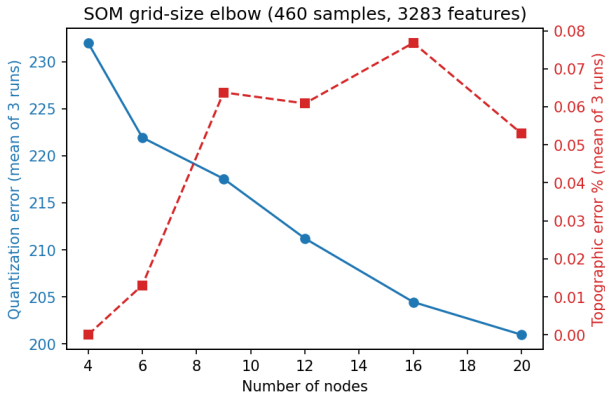
Figure 1: The SOM node weight vectors for each seasonal SOM. The same colour map is used for all seasons and for all other GHI plots.



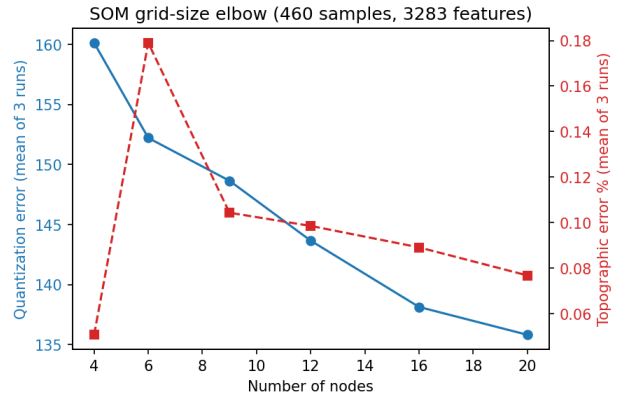
(a) Spring



(b) Summer



(c) Autumn



(d) Winter

Figure 2: SOMs were trained on different grid sizes from 2 x 2 to 4 x 5, and the final grid size chosen to minimise both topographic and quantisation error.

4 Discussion

4.1 Realism of SOM Nodes

The SOM nodes were compared to real data by computing the Pearson correlation coefficient between each real data and its best matching unit (BMU). Figure 3 shows the result of this analysis.

However, real data contains significantly more noise which are smoothed out by the nodes. Figure 4 shows a comparison of selected real data and their BMU nodes.

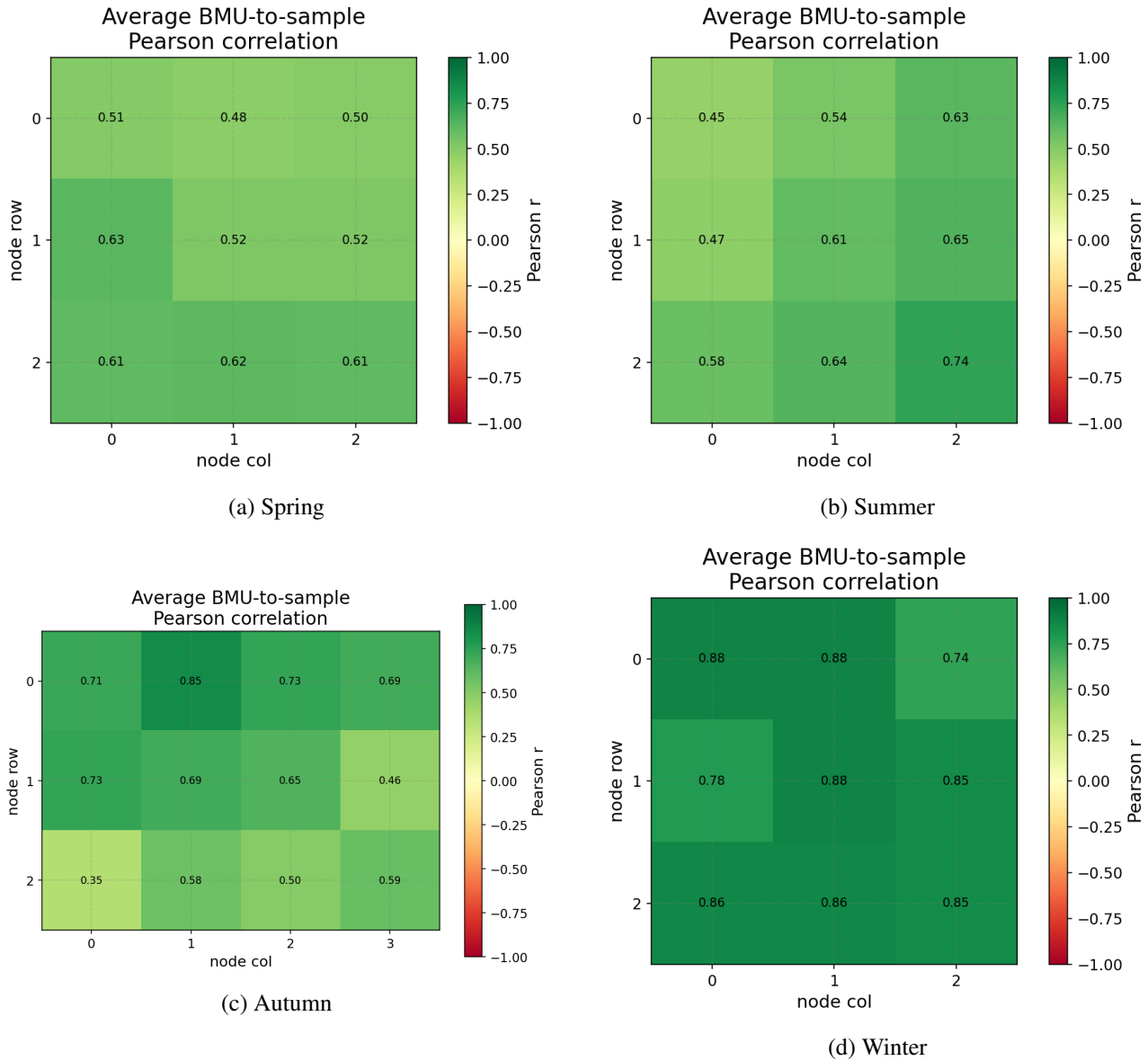


Figure 3: Correlation matrices showing the average correlation between each SOM node and all real samples assigned to it.

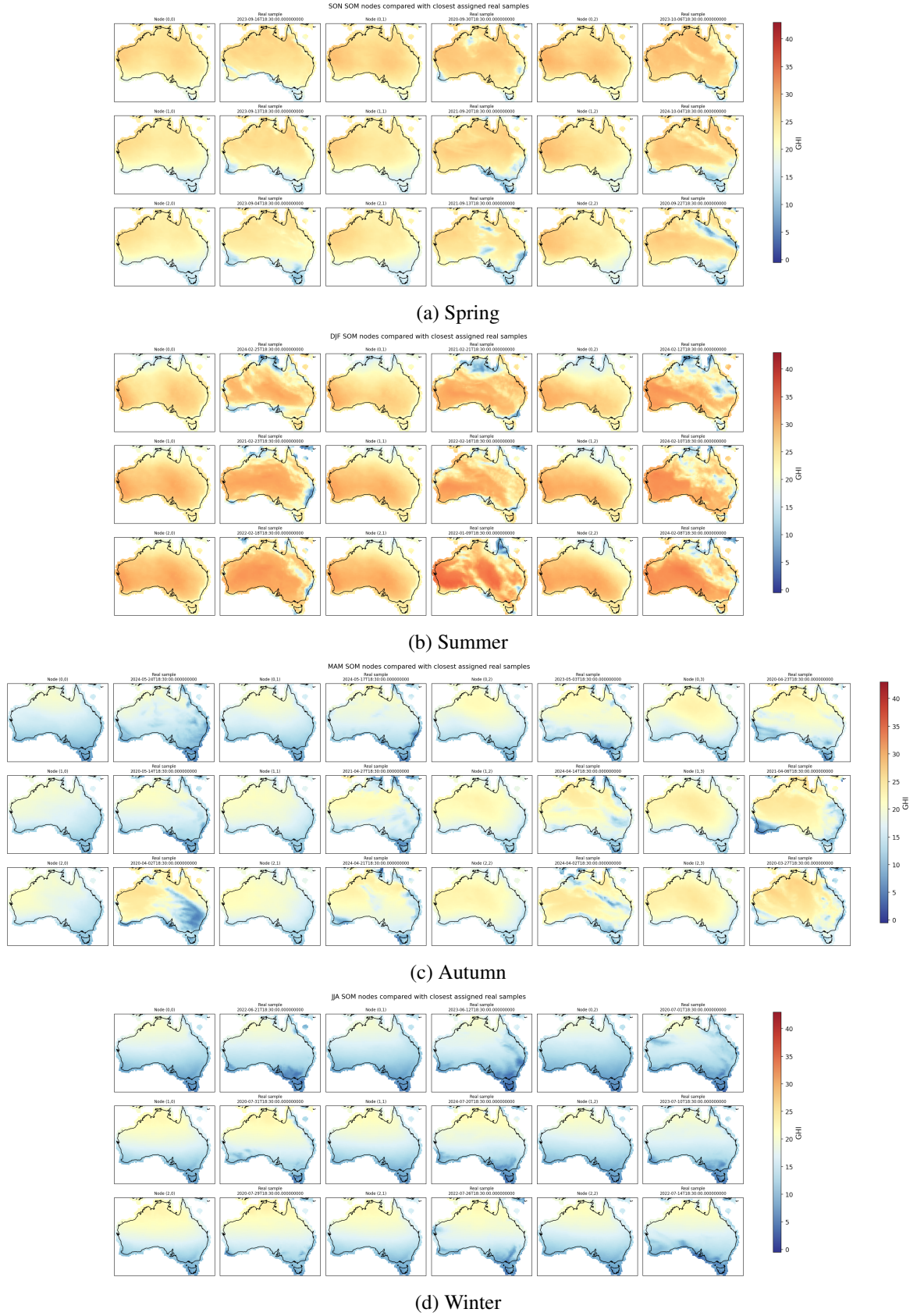


Figure 4: Side-by-side comparison of each seasonal SOM node to a randomly selected real sample which has that node as its BMU.

5 Appendix: Further Visualisations

5.1 Node Averages

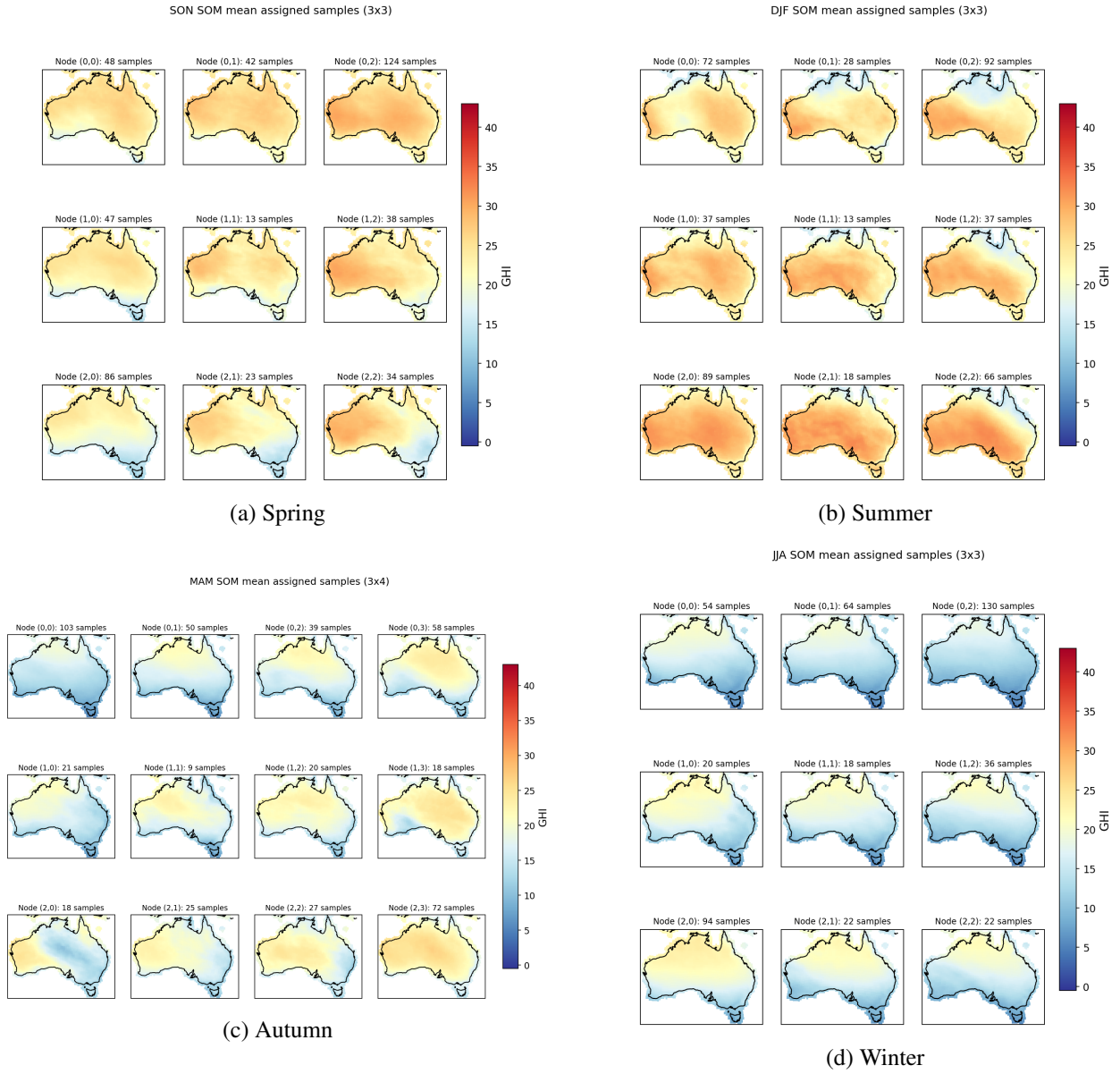


Figure 5: The average GHI of all real samples assigned to each node. This is very similar to the node weight vectors (figure 5), but with some more noise.

5.2 Correlation CDFs

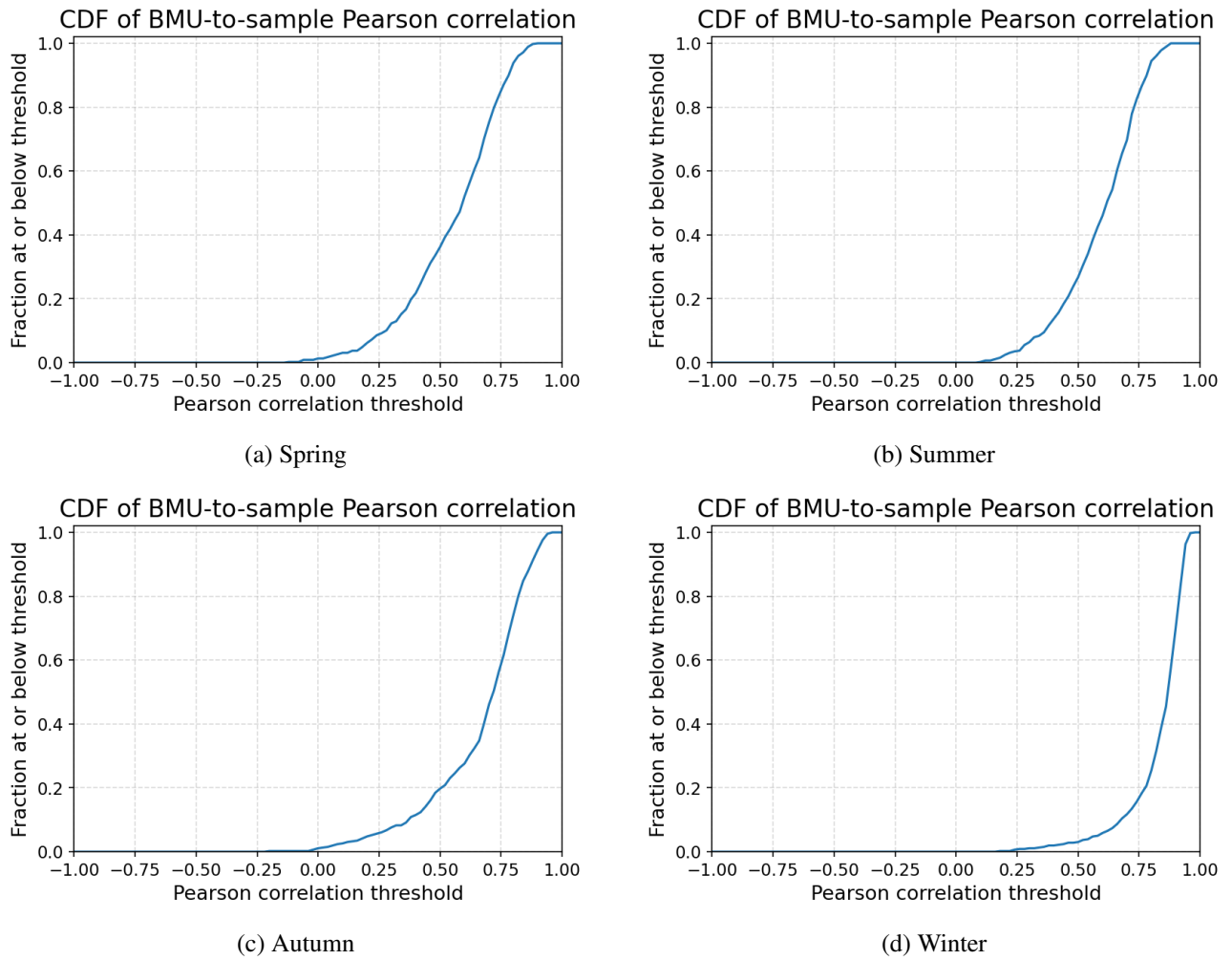


Figure 6: Correlation CDF showing the proportion of real to node correlations below each threshold.

5.3 BMU Timelines

Each date is coloured according to which node (best matching unit) it was assigned to. This plot shows how prevalent each node pattern is as well as its distribution across the training dataset. The nodes that have a large proportion of training data assigned to it appear frequently in the timeline and often in contiguous blocks.

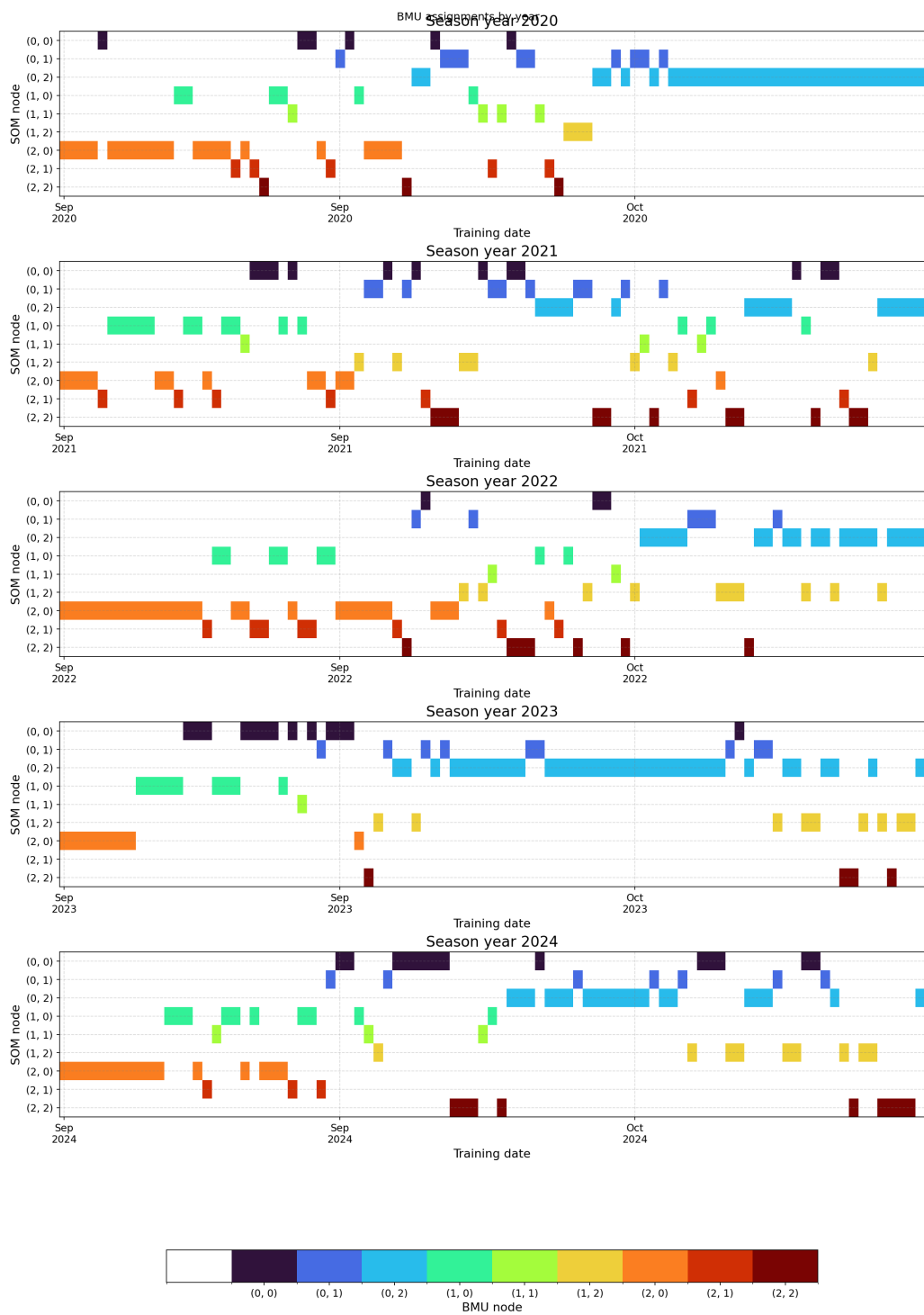


Figure 7: Spring

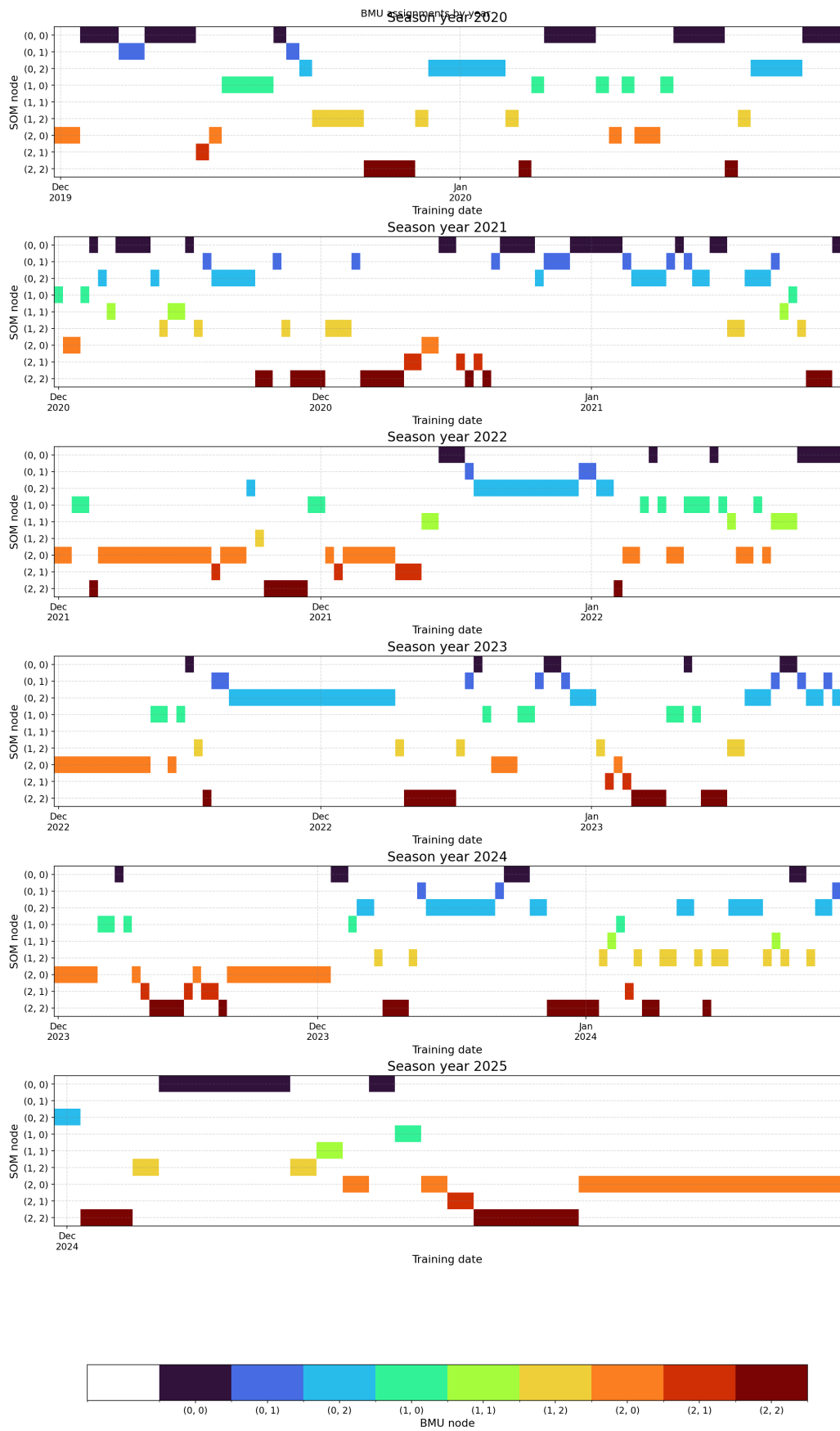


Figure 8: Summer

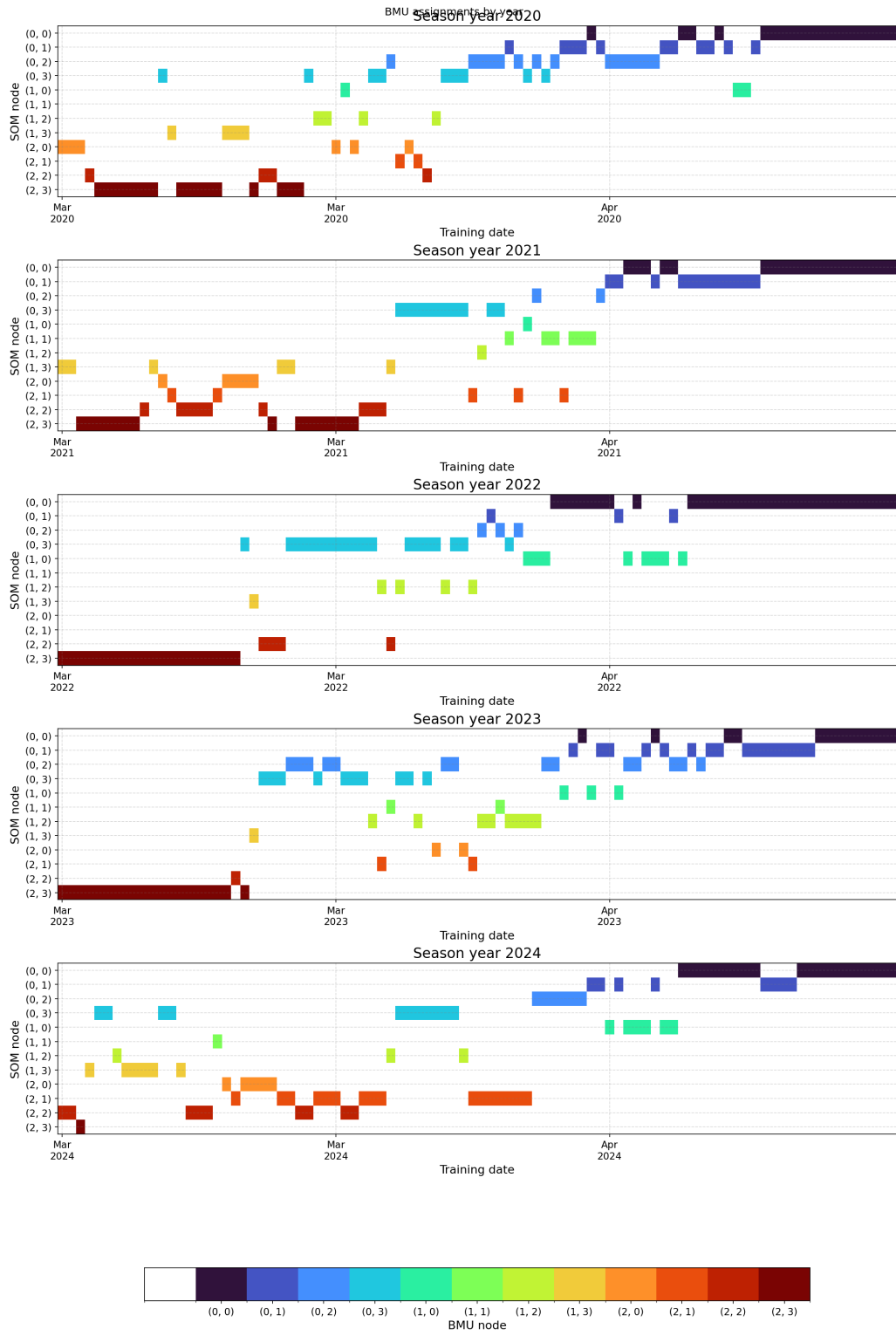


Figure 9: Autumn

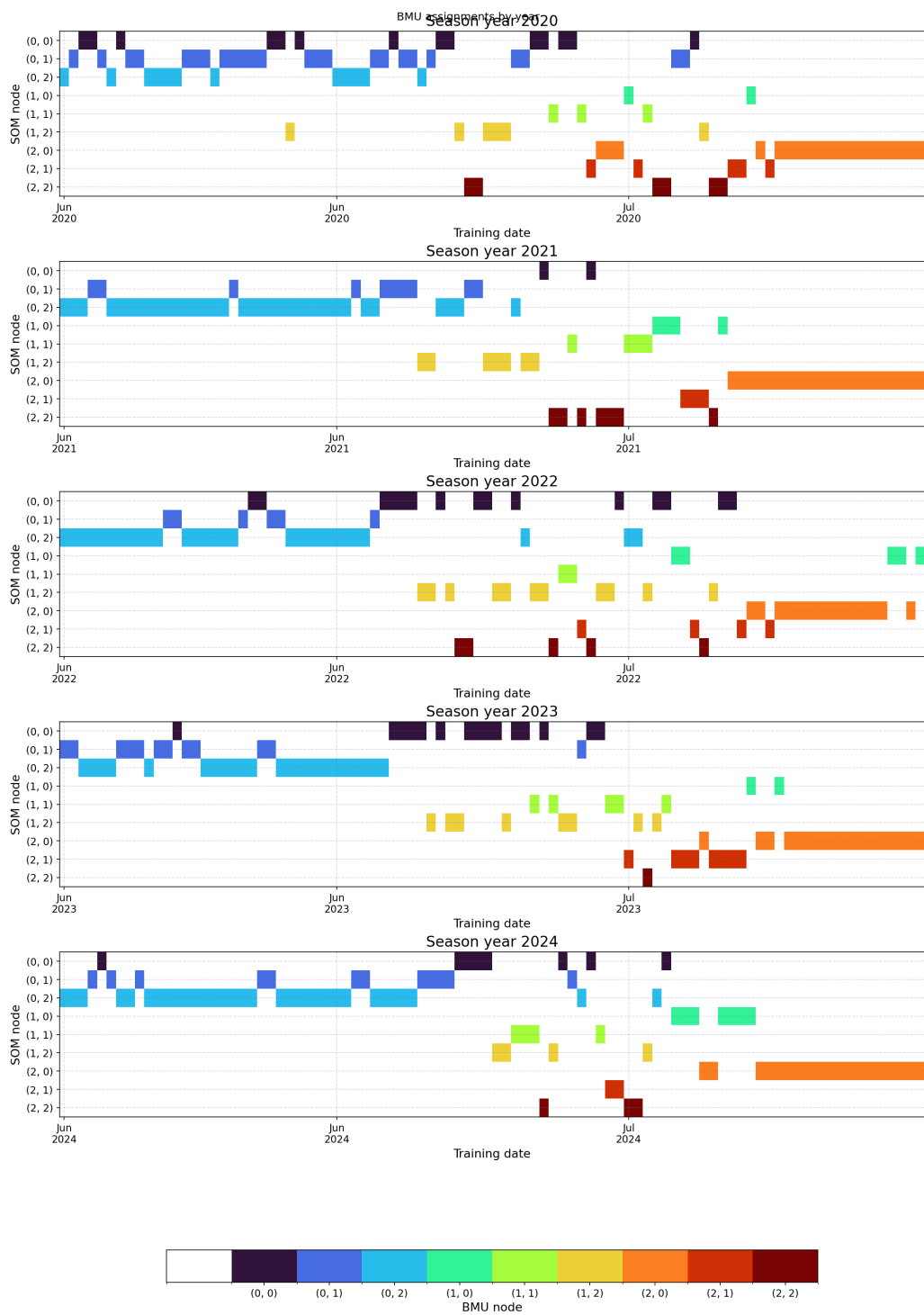


Figure 10: Winter

References

- [1] Louis Berthier. torchsom: The reference pytorch library for self-organizing maps, 2025. Documentation available at <https://opensource.michelin.io/TorchSOM/>.
- [2] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, and Eric Moulines. torchsom: The reference pytorch library for self-organizing maps, 2025. Preprint submitted to Journal of Machine Learning Research.
- [3] Peter B Gibson, Sarah E Perkins-Kirkpatrick, Petteri Uotila, Acacia S Pepler, and Lisa V Alexander. On the use of self-organizing maps for studying climate extremes. *Journal of Geophysical Research: Atmospheres*, 122(7):3891–3903, 2017.